DESCRIPTION OF COURSEWORK

Course Code	EEE308
Course Name	RF&Microwave Circuits
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2025/04
Assessment Title	Assignment : Design a Lumped and Distributed Low Pass Filter Using RF/Microwave circuit design software.

A. Introduction/ Situation/ Background Information

Microwave filters play an important role in any RF front end for the suppression of out of band signals. In the lumped and distributed form, they are extensively used for both commercial and military applications.

A filter is a reactive network that passes a desired band of frequencies while almost stopping all other bands of frequencies. The frequency that separates the transmission band from the attenuation band is called the cutoff frequency and denoted as f_c. The attenuation of the filter is denoted in decibels or nepers. A filter in general can have any number of pass bands separated by stop bands. They are mainly classified into four common types – namely low-pass, high-pass, bandpass, and band stop filters. An ideal filter should have zero insertion loss in the pass band, infinite attenuation in the stop band, and a linear phase response in the pass band. An ideal filter cannot be realizable as the response of an ideal lowpass or band pass filter is a rectangular pulse in the frequency domain. The art of filter design necessitates compromises with respect to cutoff and roll off. There are basically three methods for filter synthesis. They are the image parameter method, insertion loss method, and numerical synthesis. The image parameter method is an old and crude method, whereas the numerical method of synthesis is newer but cumbersome. The insertion loss method of filter design on the other hand is the optimum and more popular method for higher frequency applications. The filter design flow for insertion loss method is shown in Figure 1.

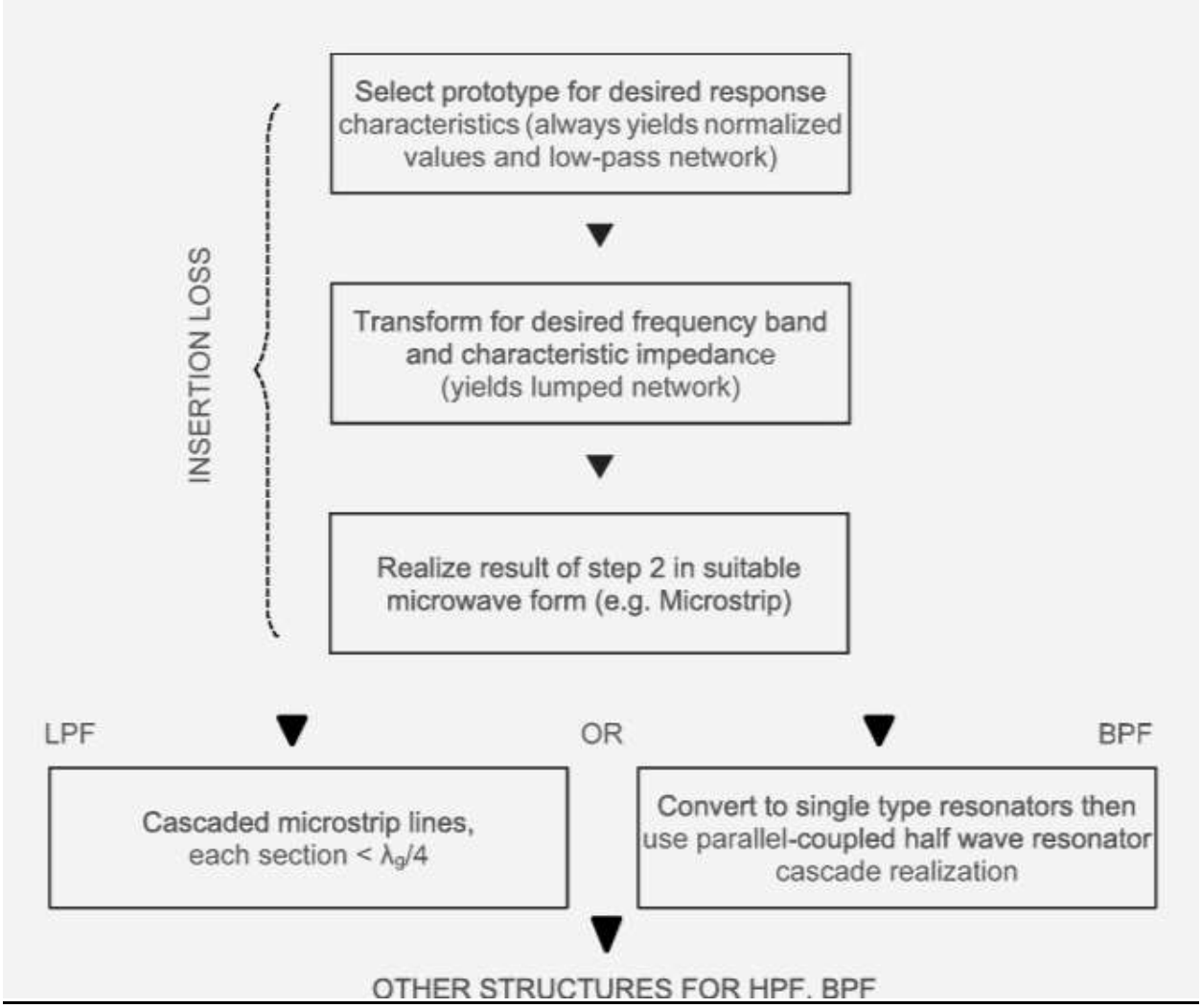


Figure 1. Filter design flow for insertion loss method

Since the characteristics of an ideal filter cannot be obtained, the goal of filter design is to approximate the ideal requirements within an acceptable tolerance. There are four types of approximations – namely Butterworth or maximally flat, Chebyshev, Bessel, and Elliptic approximations. For the prototype filters, maximally flat or Butterworth provides the flattest pass band response for a given filter order. In the Chebyshev method, sharper cutoff is achieved and the pass band response will have ripples of amplitude 1+k2. Bessel approximations are based on the Bessel function, which provides sharper cutoff, and Elliptic approximations results in pass band and stop band ripples. Depending on the application and the cost, the approximations can be chosen. The optimum filter is the Chebyshev filter with respect to response and the bill of materials. Filters can be designed both in the lumped and distributed form using the above approximations.

Design of Microwave Filters

The first step in the design of microwave filters is to select a suitable approximation of the prototype model based on the specifications. Calculate the order of the filter from the necessary roll off as per the given specifications. The order can be calculated as follows:

Butterworth Approximation

$LA(\omega') = 10 \log_{10} [1+\varepsilon(\omega' / \omega_c)^{2N})]$

where

$\varepsilon = [\text{Antilog}_{10} LA/10] - 1$ and LA = 3 dB for Butterworth

Chebyshev Approximation

$LA(\omega') = 10\log_{10} [1+\varepsilon \cos^2 [n\cos\text{-}1 (\omega' / \omega_1')]]$ when $\omega' \leq \omega 1'$

$LA(\omega') = 10\log_{10} \{1+\varepsilon\cosh^2 [n\cosh\text{-}1 (\omega' / \omega_1')]]$ when $\omega' \geq \omega 1'$

Where

ω_c is the angular cutoff frequency

ω' is the angular attenuation frequency

$LA(\omega')$ is the attenuation at ω'

N is the order of the filter

$\varepsilon = \{\text{Antilog10 } LA_r / 10\} - 1$ and LA_r is ripple in passband

The next step in the filter design is to calculate the prototype values of the filter depending on the type of approximation. The prototype values for the Chebyshev and Butterworth approximations can be calculated using the following equations:

Butterworth Approximation:

$g_0 = 1$

$g_k = 2\sin[(2k - 1)\pi / 2n]$ where $k = 1, 2, \ldots, n$

$g_{N+1} = 1$

where n is the order of the filter

Chebyshev Approximation:

$\beta = \ln(\coth LA_r\ 17.37)$

where LA_r is ripple in the passband

$\gamma = \sinh(\beta/2n)$

$a_k = \sin[(2k - 1)\pi/2n]$, $k = 1, 2, 3, \ldots, n$

$b_k = \gamma^2+\sin^2(k\pi/n)$, $k = 1, 2, 3, \ldots, n$

$g_1 = 2a_1/\gamma$

$g_k = 4a_{k-1}a_k/(b_{k-1}g_{k-1})$ $k = 2, 3, \ldots, n$

$g_{n+1} = 1$ for n odd $= \coth^2(\beta/4)$ for n even

After computing the prototype values, the prototype filter must be transformed with respect to frequency and impedance to meet the specifications. The transformations can be done using the given equations below.

For Lowpass Filter:

After impedance and frequency scaling:

$C_k' = C_k/(R_0\omega_c)$

$L_k' = R_0L_k/\omega_c$

where $R_0=50\ \Omega$

For the distributed design, the electrical length is given by:

Length of capacitance section (βl_c) : $C_k Z_l / R_0$

Length of inductance section (βl_l) : $L_k R_0 / Z_h$

Where

Z_l is the low impedance value

Z_h is the high impedance value

For bandpass filter:

After impedance and frequency scaling:

$L_1' = L_1 Z_0 \omega_0 \Delta$ $C_1' = \Delta L_1 Z_0 \omega_0 / L_2' = \Delta Z_0 / \omega_0 C_2$

$C_2' = C_2 Z_0 \Delta \omega_0 / L_3' = L_3 Z_0 / \omega_0 \Delta$ $C_3' = \Delta L_3 Z_0 \omega_0$

Where

Δ is the fractional bandwidth $\Delta = (\omega_2 - \omega_1) / \omega_0$

Objective:

Use a Microwave Circuit Design software to:

(a) Design and simulate a 4th order Lumped Lowpass Filter operating between 0.1 GHz and 5GHz using Butterworth approximation for $Z_l = 8\ \Omega$ and $Z_h = 95\ \Omega$

(b) Design a stepped-impedance type Butterworth low pass filter (Example 16-2-1of the GW-Instek application note) with cutoff frequency at 2GHz and > 20dB attenuation at 2.8 GHz. The input and output impedance are both 50 Ω . The PCB material is FR4 with dielectric constant 3.8 and thickness 1.18 mm.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Build(design) microwave circuits and measure(lab) its signals

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

The reports must include abstract ,introduction, theory, methodology, results &discussion, conclusion and references.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Design a Lumped and Distributed Low Pass Filter Using RF/Microwave circuit design software.	100 %
	TOTAL	**10%**

F. Tasks

Design a Lumped and Distributed Low Pass Filter Using RF/Microwave circuit design software. The assignment report must have the components outlined in the rubrics in Appendix 1 and submitted before 6 June 2025.

APPENDIX 1

MARKING RUBRICS

	Excellent	Good	Adequate	Poor	Lacking
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.

methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual.	The description gives generalities, enough for reader to understand how the experiment was conducted	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted	Part is missing or plagiarized
Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.

Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or realworld connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References Formatting	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.

Component Title	ASSIGNMENT				Percentage (%)	100
Criteria	Score and Descriptors					Weight (%) Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)	
Abstract						10
Introduction & Theory						10
Methodology/ Experimental						35
Results, Disscussion Conclusion						40
Formatting References						5
					TOTAL	100